MIL-C-7460A(USAF)
4 February 1971

SUPERSEDING
MIL-C-7460(USAF)
7 February 1952

MILITARY SPECIFICATION

CHROMIUM PLATING, POROUS CHANNEL TYPE, AIRCRAFT ENGINE CYLINDERS, GENERAL SPECIFICATION FOR

1. SCOPE

1.1 Scope. This specification covers the requirements for porous (channel type) chromium plating the cylinder bore of aircraft engine cylinders.

2. APPLICABLE DOCUMENTS

* 2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

TT-A-468	Aluminum-Pigment; Powder And Paste, For Paint
TT-E-489	Enamel, Alkyd, Gloss (For Exterior And Interior Surfaces)
TT-N-97	Naphtha; Aromatic
TT-V-109	Varnish, Spar, Alkyd-Resin

Military

MIL-C-16173	Corrosion Preventive Compound, Solvent Cutback, Cold-Application
MIL-E-25111	Engines, Aircraft, Reciprocating, Qualification Tests For

STANDARDS

Military

MIL-STD-143	Standards And Specifications, Order Of Precedence For The Selection Of

PUBLICATIONS

Air Force

T.O. 2R-1-17	Reconditioning And Use Of Steel Barrel And Chromium - Plated Engine Cylinders - All Reciprocating Engines

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 Materials. All materials used in the process of chromium plating shall be entirely suitable for electro-deposition of high quality coatings for aircraft engine use.

* 3.1.1 Selection of materials. Specifications and standards for all materials, parts, and Government certification and approval of processes and equipment, which are not specifically designated herein and which are necessary for the execution of this specification, shall be selected in accordance with MIL-STD-143. When contractor's specifications are used for materials and processes which affect performance or durability of the finished product, such specifications shall be subject to release by the Government.

3.2 Electrochemical process. Chromium deposition shall be out of a chromic acid type bath, of composition and with operating conditions as necessary to produce the type of chromium plating defined in this specification.

* 3.3 Cylinder conditioning prior to plating. Cylinder grinding or other surface preparation prior to plating shall be accomplished by the contractor unless otherwise specified in the contract. When necessary equipment is available at the plating contractor's facility, the grinding or other surface preparation shall be accomplished immediately prior to plating. When grinding is accomplished at a source other than the plating facility or whenever the cylinder cannot be plated immediately following the grinding operation or other surface preparation, the cylinders shall be prepared for shipment and storage in a manner which will insure adequate protection against corrosion, deterioration or physical damage until next step in cylinder conditioning and plating can be accomplished.

3.3.1 Cylinders shall be ground or otherwise taken to an oversize bore diameter not less than 0.010 nor more than 0.025 inches larger than the nominal bore diameter for each type cylinder or to limits as otherwise specified by the Procuring Agency. The nominal bore diameter shall be as specified on the engine contractor's drawing of the cylinder machining and studs assembly.

3.3.2 The cylinder shall have a finish of 16 micro inches root mean square or less.

3.3.3 The cylinders shall be finish machined to close limits on taper and out-of-round such that the requirements of this specification in regard to taper, plating thickness, and out-of-round for the finished chromium plated cylinder shall be met.

3.3.4 The cylinder or sleeve will be entirely suitable for engine installation except for plating in conformance with this specification.

3.3.5 All aircraft cylinder assemblies, particularly the cylinder bore, shall be free of all residue affecting the quality of the plating application. The contractor shall be responsible for inspecting the cylinder for all defects which would affect its serviceability after plating.

* 3.3.6 Disposition of rejected cylinders. Cylinders rejected for any reason shall have all finished and unpainted surfaces covered with a corrosion-preventive compound conforming to MIL-C-16173, Grade 1. These cylinders shall be set aside and disposition shall be made in accordance with instructions from the procuring agency.

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3.4 Plating.

3.4.1 Plating requirements. The application of chromium deposition shall be a continuous operation without interruption of the electric current. Any deviation from this practice must have written approval from the procuring agency. The chromium deposit shall be homogeneous, uniform in appearance, and free from pits, nodules, blisters and defects except as specifically permitted by this specification. Double plating will be permitted only by written approval of the details of the process by the procuring agency. Interruption of the deposition of chromium to measure the diameter of the cylinder or sleeve to determine the amount of chromium deposited will be permitted providing subsequent chromium deposits will be removed by the post treatment and finishing operations.

3.4.2 Excess deposits. Chromium deposition shall not be evident on any part of the cylinder other than on the cylinder bore. Chromium shall not be allowed to project beyond either the head or the skirt end of the barrel. Any extraneous chromium deposition such as on valve guides, valve seats, or the cylinder head shall be removed by means acceptable to the procuring agency.

3.4.3 Nondeposition of chromium. Nondeposition of chromium on the last 0.062 inch of the bore at the head end of the cylinder shall be permitted. Nondeposition of chromium at the skirt end of the cylinder shall not exceed 0.031 inch.

3.4.4 Etching effects. Etching shall not be evident on any component part of the cylinder, except for the cylinder bore.

3.4.5 Sharp corners. The edge of the plating at the skirt end of the cylinder shall be radiused either by means of the plating technique or by mechanical means to provide an approximate 1/32 inch radius to prevent damage to piston rings on installation. At the head end of the cylinder the area not exceeding more than the top two thirds of the top ring travel clearance shall be relieved with a bevel not to exceed 5 degrees from the center line of the cylinder and shall be accomplished by means acceptable to the procuring agency.

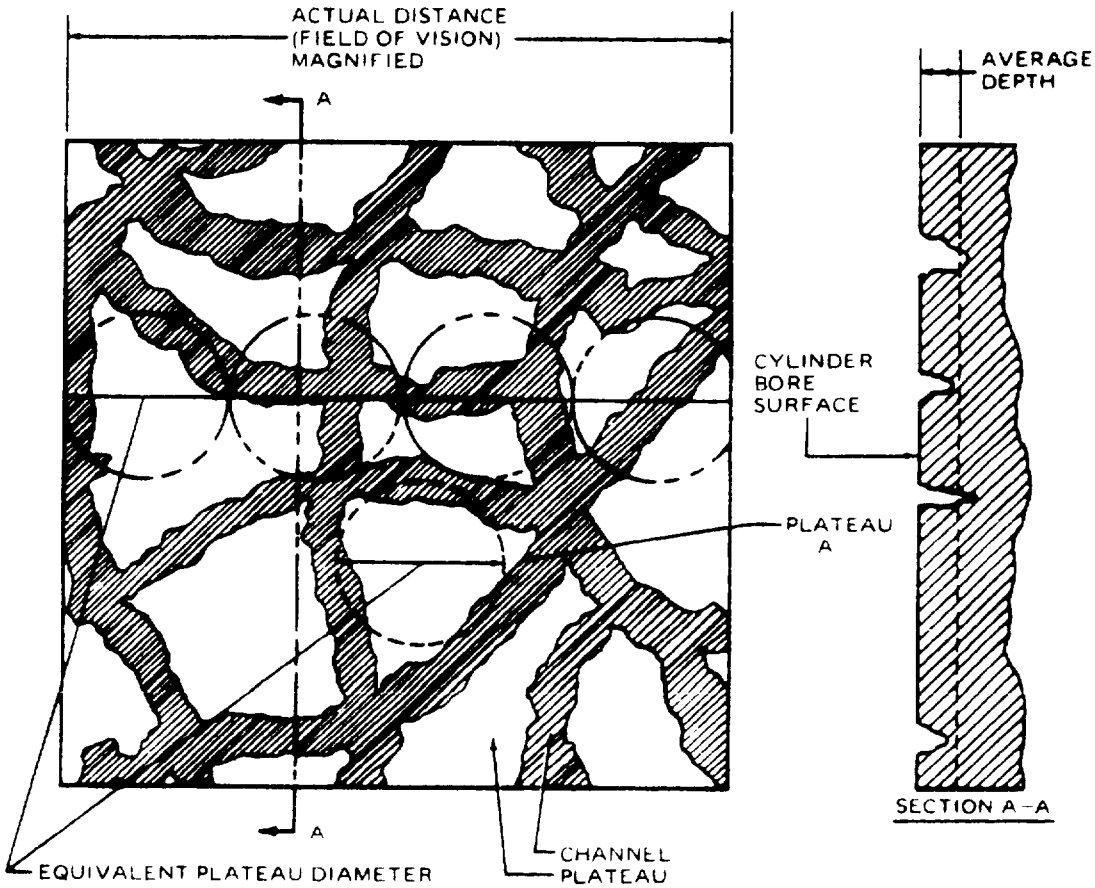
3.4.6 Adherence. There shall be no evidence of separation of the chromium from the base material at the interface or separation of the second layer of chromium from the first layer of chromium at the interface, in the case of double plating.

3.4.7 Plateaus. The size of the plateaus and the depth and width of the channels, as shown in Figure 1, is dependent upon the plating, etching and honing conditions. It is desirable to maintain the average plateau size to approximately 0.020 inch equivalent diameter. The average plateau size shall not be more than 0.035 inch equivalent diameter and shall not be less than 0.010 inch equivalent diameter.

3.4.8 Porosity.

3.4.8.1 Maximum depth of porosity. A maximum depth of porosity is desirable after honing the cylinder to a satisfactory bearing surface. To arrive at this maximum depth of porosity, all cylinders produced in the contractor's plant shall have continuous channels throughout the ring travel area of the cylinder bore.

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AVERAGE PLATEAU SIZE DETERMINATION

A. SELECT A PLATEAU OF AVERAGE SIZE
B. COUNT THE NUMBER OF PLATEAUS OF THE SIZE SELECTED ABOVE WHICH COULD BE PLACED IN A STRAIGHT LINE ACROSS THE FIELD OF VISION.
C. DIVIDE THE ACTUAL DISTANCE COVERED BY FIELD OF VISION BY NUMBER OF PLATEAUS DETERMINED UNDER "B". THE RESULT OF WHICH WILL BE CALLED THE AVERAGE EQUIVALENT PLATEAU DIAMETER

FIGURE 1

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3.4.8.2 Percent of porosity. The percentage reduction of the cylinder bore surface area after plating, occasioned by the channels or grooves, shall be 25 to 40 percent anywhere within the total ring travel area. The percentage porosity may exceed these values in the surface areas outside the ring travel area.

3.4.9 Pitting. Pitting is undesirable. Pitting, if present, shall be allowed subject to the following limits:

3.4.9.1 No pit shall be larger than 0.032 inch in diameter.

3.4.9.2 Not more than 2 pits 0.032 inches in diameter nor more than four pits averaging 0.015 inch in diameter shall be allowed in a 1/4 inch radius.

3.4.9.3 Not more than 50 pits averaging 0.015 inch in diameter nor more than 25 pits averaging 0.032 inch in diameter shall be allowed in any cylinder.

3.5 Finish.

* 3.5.1 Surface finish. The cylinder bore shall have a honed finish with a surface smoothness on the plateau surface equal to or better than that obtained on the first article inspection samples. The first article inspection samples to be used for inspection purposes shall consist of one or more of the cylinders submitted by the contractor for the first article inspection. At the request of the contractor the procuring agency will furnish the data necessary for the contractor to finish hone the first article inspection samples.

3.5.1.1 Honing area. No unhoned surfaces shall be permitted within the piston ring travel area. Partially honed areas shall be permitted in any portion of the cylinder bore provided the percentage reduction of area and other requirements are within limits. The portion of the cylinder barrel bore surface beyond the extremities of the travel of the piston rings need not be honed. In no case, however, shall the dimensions of these areas be such as to interfere with the operation of the piston.

3.5.2 Cleaning. The chromium plated surface of the entire cylinder bore, including the grooves and the channels between the plateaus, shall be thoroughly cleaned of all debris by a method acceptable to the procuring agency. The cleaning operation shall be accomplished immediately following the final honing operation.

3.6 Cylinder dimensions after plating and honing. After plating and honing to a satisfactory surface bearing condition, as specified in applicable portions of this specification, the cylinder bore shall conform to the following:

3.6.1 Depth of chromium. The thickness of the chromium on the finished bore shall not be less than 0.0035 inch.

3.6.2 Bore diameter. The average nominal bore diameter shall be within minus 0.0015 inch to plus 0.0025 inch of the nominal bore diameter shown by the engine manufacturer's drawing of the cylinder or as otherwise specified by the procuring agency.

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3.6.3 Axis squareness limits. The axis of the final chromium plated bore shall be held within the required squareness limits with respect to the cylinder flange mounting face as specified on the engine manufacturer's drawing or as otherwise specified by the procuring agency.

3.6.4 Taper. For straight bore cylinders and for the straight section of choked bore cylinders, positive cylinder taper shall not exceed 0.003 inch on the diameter and negative taper shall not exceed 0.002 inch on the diameter for the entire length of the straight section.

3.6.4.1 Rate-of-taper. The rate-of-taper shall not exceed 0.001 inch per inch from the skirt end of radial aircraft engine cylinders to a point approximately one inch beyond the side of the hold-down flange nearest the cylinder head. The rate-of-taper shall not exceed 0.0005 inch per inch from a point one inch beyond the hold-down flange to the beginning of the choke. In the case of cylinder sleeves the rate-of-taper shall not exceed 0.0005 inch per inch for the entire length of the bore.

3.6.5 Choke (engine manufacturer's drawings). A positive choke of 0.004 inch will be allowed on radial aircraft engine cylinders which are normally to be plated straight. Those radial aircraft engine cylinders which are normally to be plated with a choke or those cylinders which incorporate a specified amount of choke before plating shall have a choke not to exceed the maximum as specified by the engine manufacturer's drawings of the cylinder machining and studs assembly and a minimum choke which shall not be less than 0.004 inch. Unless otherwise specified, no choke shall be permitted in cylinders or sleeves where the engine manufacturer's drawing does not specify a choke.

3.6.5.1 Rate-of-choke. The maximum permissible uniform rate-of-choke per inch will be determined by dividing by two the maximum allowable choke as determined from the engine manufacturer's drawing. This uniform rate-of-choke is also applicable to the head and ring travel clearance area.

3.6.6 Choke (cylinder contour graphs). In lieu of 3.6.5 and 3.6.5.1, and when so specified in the contract, the choke and rate-of-choke shall be as specified on the cylinder bore contour graphs, for choke bore cylinders. These contour graphs will be furnished by the procuring agency.

3.6.7 Out-of-round. Out-of-round limits shall not exceed 0.003 inch for the area swept by the piston rings.

3.7 Exterior finish.

* 3.7.1 Painting. The exterior surface of the steel flange and barrel of each cylinder shall be painted to insure complete coverage with spar varnish in accordance with TT-V-109, pigmented with aluminum paste (12 ounces per gallon), TT-A-468, Type II. Aromatic naphtha, in accordance with TT-N-97 shall be used as a thinner, when required, to obtain proper consistency of the mixture. The area painted shall begin with the outside edge of the flange and continue to the bottom of the head for cylinders having steel fins. Care shall be exercised to avoid paint deposition on flange mounting surface, stud holes, and centering washer seats.

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* 3.7.2 Identification marking. Chromium plated cylinders shall be identified by painting the exterior of the cylinder from the lower corner of the outside edge of flange to the bottom cooling fin on the barrel with International Orange light stable molybdate enamel, TT-E-489. Care must be taken to assure that paint is not applied to the flange mounting surface or the hold down stud centering washer seats.

3.7.3 Etching. Each cylinder shall be electrically etched on the barrel skirt diametrically opposite any existing historical data with the words "Chrome Plated" and a code marking. Care shall be exercised to assure legibility and to avoid deep, heavily burnt impressions. Upper case letters approximately 1/4 inch in height shall be used.

* 3.7.4 Code marking. The code marking shall be placed directly below the words "Chrome Plated" and inscribed in the order indicated.

a. Arabic numeral indicating the month in which the cylinder was plated.

b. Dash.

c. The last two digits of the year in which the cylinder was plated.

d. Dash.

e. Code letters (T.O. 2R-1-17) designating the contractor plating the cylinder.

f. Dash.

g. Arabic numeral indicating the monthly plating serial number assigned by the plating contractor and shall be determined as follows: Each month a separate series beginning with one shall be used for cylinders from each type of engine.

h. Dash.

i. Arabic numerals indicating the amount of oversize in thousandths of an inch of the cylinder before plating.

j. Example: "8-50-XXX-4-15" has the following meaning: Subject cylinder was plated in August 1950 by XXX Company (T.O. 2R-1-17) and was the fourth cylinder plated by that particular company that month. The cylinder was 15 thousandths oversize before plating.

3.7.4.1 Report designation. Any reports pertaining to specific cylinders shall be reported by this number with the engine type prefixed thereto. Example: "R-2000-8-50-XXX-4-15."

3.8 Drawings. The requirements of the particular engine manufacturer's drawings, relative to cylinder bore dimensions, are applicable as detail requirements of this specification except as otherwise specified herein.

* 3.9 Workmanship. The workmanship and finish on all chromium plated cylinders shall be in accordance with high grade aircraft engine practice.

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4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 Classification of tests. The inspection and testing of chromium plated cylinders shall be classified as follows:

a. First article inspections are those tests accomplished on samples representative of the production of porous chromium plated cylinders to determine that the production of porous chromium plated cylinders meets the requirements of this specification. The first article inspections shall be satisfactorily completed prior to the acceptance of any chromium plated cylinders on the contract.

b. Quality conformance inspections are those tests to which porous chromium plated cylinders are subjected during or after completion of manfacture to determine whether the specification requirements have been met and the article is satisfactory. The quality conformance inspections shall be conducted at the contractor's facility.

* 4.3 First article inspection. The first article inspection shall be conducted by the procuring agency and shall consist of the inspection tests herein and a 150 hour endurance test in accordance with MIL-E-25111. At the option of the procuring agency and where such 150 hour endurance tests have previously been accomplished by the Government on specific engine types and models, shorter endurance tests may be conducted by the procuring agency in accordance with 4.3.4.

* 4.3.1 First article inspection samples. The first article inspection samples shall consist of the first Government furnished engine cylinders, having the bores chromium plated and finished, by each plating contractor, for each engine type and of a quantity necessary to equip completely one engine, and in addition, 50 percent spare cylinders. Samples shall be forwarded to the destination specified by the procuring agency, plainly identified by securely attached durable tags, marked with the following information:

For First Article Inspection
Specification MIL-C-7460A(USAF)
Cylinder - Engine Type and Model
Contract or Order No. _____
(Plating Mfr's Name or Trademark)
Submitted by (Name) (Date)

* 4.3.2 Process report. Contractors shall furnish a process report, certified by affidavit, giving the details of the process used to meet the requirements of this specification. This report shall be furnished with or prior to submission of the first article inspection samples.

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4.3.2.1 Description. A complete description of the chromium plating process utilized, including data relative to the following:

a. Preparation of cylinder for plating.

b. Details of waxing methods, including trimming of wax.

c. Details of special precautions taken to insure uniform deposit of chromium at both ends of the cylinder bore and steps taken to prevent plating and etching of the head and head recess at the barrel edge.

d. Location, dimensions and special contours of anode.

e. Details of etching prior to plating, including time, temperature, density of electric current, and bath analysis.

4.3.2.2 Plating. Details, and variations permitted in bath analysis, density of electric current and temperature.

4.3.2.2.1 Average plating efficiency.

4.3.2.3 Post treatment. Details of and variations permitted in post treatment, including time, bath analysis, temperature and density of electric current.

4.3.2.4 Honing. Details of honing procedure including coolant, type of honing stones, processes used, and particular techniques developed.

4.3.2.4.1 Average honing time per cylinder.

4.3.2.5 Cleaning. Details of method used to clean debris from porosity.

* 4.3.3 Test conditions. The chromium plated cylinders will be assembled on the engine, at the engine testing facility specified by the procuring agency, for first article inspection running.

4.3.4 Tests.

4.3.4.1 Block testing. The test engine will be block tested as follows:

4.3.4.1.1 Preliminary run. The nature and extent of the preliminary running period before the final run will be determined by the procuring agency. The maximum run-in time for any engine will not exceed seven hours, or as otherwise authorized by the procuring agency.

4.3.4.1.2 Final run. The engine will be subjected to a two hour run at normal rated power and speed, during which time an accurate check of oil consumption will be made and recorded. Excessive oil consumption beyond the limits as specified in the applicable engine model specification can be cause for rejection of cylinders at the discretion of the procuring agency where it can be established that the quality or dimensions of the chromium plating has contributed to the excessive oil consumption.

4.3.4.1.3 Inspection after final run. Upon completion of the two hour final run, all cylinders will be removed for complete visual and dimensional inspection. If there is any evidence of scuffing, excessive wear of rings or cylinders, or scratching due to honing debris remaining in the porosity, the cylinders shall be rejected.

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* 4.3.4.2 Rejection. No further first article testing of cylinder will be done following a rejection until suitable corrective action has been taken by the contractor. Such corrective action shall be subject to the approval of the procuring agency.

* 4.4 Quality conformance inspection. The finished plated engine cylinder and all plating materials entering into the processing thereof, shall be subject to inspection by authorized Government Inspectors. Inspection of each of the finished chromium plated cylinders shall cover the following:

a. Workmanship as defined in 3.9.

b. Conformance to drawings or dimensional requirements and compliance with applicable specifications including this specification and those requirements specified by the procuring agency.

4.4.1 Inspection for workmanship. Each chromium plated cylinder shall be inspected after finishing to determine conformance with the requirements of this specification with respect to material and workmanship. Visual inspection shall be accomplished with the aid of a suitable 50 power microscope.

4.4.2 Dimensional inspection. The thickness of chromium deposit and finished dimensional requirements shall be determined by bore measurements taken before plating and after finish honing, with a suitable dial indicating micrometer.

4.4.3 Plateau size. The size of the plateaus shall be determined by the use of a suitable type of 50 power microscope unless otherwise approved by the procuring agency.

4.4.4 Percent porosity. The percentage reduction of cylinder bore surface area shall be estimated by visual examination by means of the microscope specified in 4.4.3, and with the aid of sample 50 power photographs on which the reduction of surface area has been calculated. These sample photographs will be furnished by the procuring agency.

4.4.5 Surface finish of plateaus. The surface finish of the plateaus on a finished chromium plated cylinder bore shall be inspected by methods and with instruments approved by the procuring agency.

4.5 General.

* 4.5.1 Changes in material or process. No changes shall be made in the materials or processes used in plating engine cylinders which have successfully completed the first article inspections, except when such changes are approved by the procuring agency. Such changes when submitted to the procuring agency shall be accompanied by substantiating information clearly illustrating the proposed change.

4.5.2 Approval of changes. Approval of changes from the qualified materials and processes to be incorporated in the production of cylinders does not relieve the contractor of full responsibility for the results of such changes on any engine cylinder characteristics.

4.5.3 Acceptance or approval of material or process reports by the procuring agency shall in no case be construed as a guarantee of the acceptance of the finished product.

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4.5.4 Materials test. When required, samples of materials used in the chromium plating process shall be selected in the manner and quantity, and subjected to the tests specified in the detail materials specification. Samples of materials and parts required for tests shall be in addition to the order and shall be furnished free of charge to the procuring agency.

5. PREPARATION FOR DELIVERY

* 5.1 Packaging, packing and marking requirements for the finished aircraft engine cylinder shall be as specified by the procuring agency.

6. NOTES

6.1 Use. Engine cylinders having their bores chromium plated and fulfilling the requirements set forth in this specification, are intended for installation on aircraft engines undergoing major overhaul.

6.2 Definitions.

6.2.2 Cylinder taper. Cylinder taper is defined as the difference between successive measurements of bore diameter made in any one plane which includes the cylinder axis.

6.2.3 Rate-of-taper. The rate of change in diametrical dimensions per unit of length is indicated between successive measurements made in any one plane which includes the cylinder axis.

6.2.4 Cylinder choke. A gradual reduction in cylinder bore diameter starting approximately 2-1/2 inches from the head end of the cylinder barrel, resulting in a positive taper or constriction, to the upper end of the cylinder. Choke is designed into the head end of the cylinder bore, while taper, as in the straight section of the cylinder, is a manufacturing tolerance on a nominal dimension.

6.2.5 Rate-of-choke. The amount of constriction of the bore diameter per unit of length at the head end of the cylinder barrel.

6.2.6 Head end ring travel clearance. The distance between the bottom edge of the lowest piston ring at the bottom dead center position of the piston and the extremity of the cylinder bore at the head end of the cylinder.

6.2.7 Skirt end ring travel clearance. The distance between the bottom edge of the lowest piston ring at the bottom dead center position of the piston and the extremity of the cylinder bore at the skirt end of the cylinder.

6.2.8 Choked cylinder. Cylinders in which a section at the head end of the bore is deliberately choked or constricted so that the diameter is reduced more than 0.004 inches.

6.2.9 Straight cylinders. Cylinders in which the entire bore length is made straight or where the total choke does not exceed 0.004 inch.

6.2.10 Cylinder out-of-round. The difference between maximum and minimum diameter measurements made in any one plane normal to the cylinder axis.

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6.2.11 Pits. Indentations or depressions in the plating which may be the result of a surface flaw in the base metal or of various factors in the plating process.

6.2.12 Interface. The surface over which a layer of chromium is deposited.

6.2.13 Interrupted channels. Primary channels which are not continuous and do not isolate the individual plateaus necessary for satisfactory operation.

6.2.14 Ring travel area. The surface area in the cylinder bore swept by the piston rings.

6.2.15 Plateau equivalent diameter. The diameter of a circle whose area equals that of the plateau in question.

6.2.16 Partially honed area. An area where only portions of the individual plateaus have been completely finished to a 100 percent honed surface.

6.2.17 Percent porosity. Percent porosity is the ratio of the area of the channels to the total surface area considering a unit surface area and is expressed as a percent. The mathematical formula for percent porosity is:

$$\text{Percent porosity} = \frac{\text{area of channels}}{\text{total area}} \times 100 \text{ (for a unit area under consideration)}$$

6.3 Ordering data. Invitation for bids, contracts and purchase orders should state the conditions for the following:

* 6.3.1 First article inspection. It is expected that the contract or purchase order will specify that a sufficient number of cylinders to equip one engine of the type and model designated plus 50 percent spares will be required as first article inspection samples and that these first article inspection samples will be subjected to the first article inspections to determine compliance with the requirements of this specification. Invitation for bids and the contract should specify where the first article inspection will be accomplished and the point of inspection for these tests.

* 6.4 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodian:
Air Force - 82

Review Activity:
Air Force - 82

Preparing Activity:
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Project No. MFFP-F048

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 68 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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DEPARTMENT OF THE AIR FORCE

OFFICIAL BUSINESS

SAN ANTONIO AIR MATERIEL AREA
SERVICE ENGINEERING DIVISION (MMTOP)
KELLY AFB, TEXAS 78241

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